For Examiner's Use

2 A satellite of mass 1 200 kg is launched from the Earth's surface into a circular orbit at a height of 35 600 km directly above the equator. The radius of the Earth is 6 400 km.

Fig. 2.1 shows the variation with distance from Earth's centre r of gravitational potential Φ due to Earth.

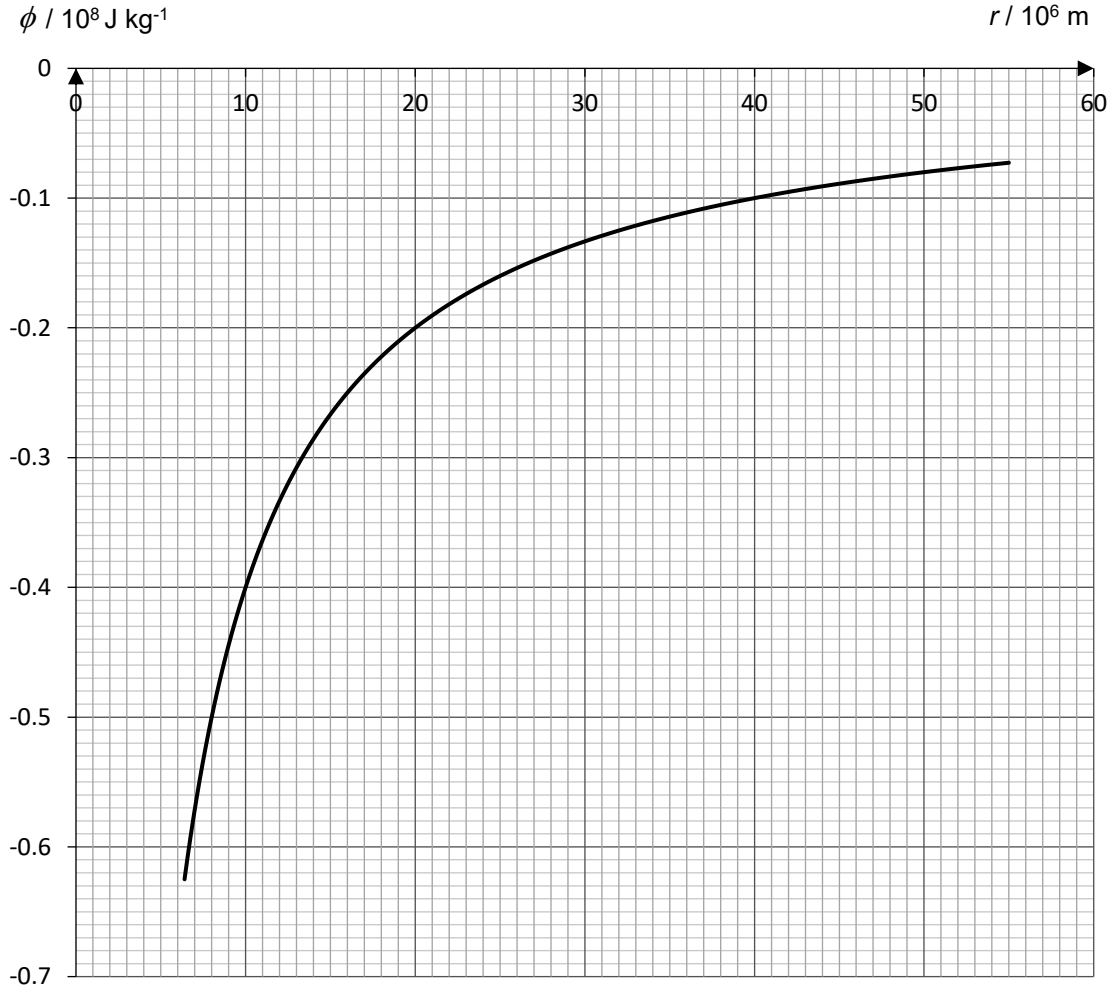

Fig. 2.1

(a) Explain why the gravitational potential graph is always negative.

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……

…… [2]

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(b) Using Fig. 2.1, show that the gravitational force acting on the satellite by the Earth when it is in orbit is about 270 N.

[3]

(c) Hence or otherwise, calculate the kinetic energy of the satellite when it is at the orbit.

kinetic energy = ... J [3]

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(d) Use Fig. 2.1 and your answer in **(c)** to determine the minimum amount of energy required to launch the satellite to the orbit. Assume that the kinetic energy of the satellite at the surface of Earth is 0 J.

energy = ……. J [3]

(e) The actual energy required to send the satellite to the orbit is different from the energy calculated in **(d)**.

Suggest a reason why.

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…… [1]